$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k}$: UV divergence in $\overline{\mathrm{MS}}$ and logarithms of the longitudinal cutoff Λ

Summary

- The only UV divergence is at $z \to x$. Its coefficient is the gluon splitting function $\mathcal{P}(\xi) = \frac{\xi}{1-\xi} + \frac{1-\xi}{\xi} + \xi(1-\xi)$, $\xi = p^+/k^+$, and the $ffU(x)U(z)$ term does not change it. It is removed in $d_\perp = 2-\epsilon$ with $\overline{\mathrm{MS}}$ (Steps 1–4); the result is eq. (21).
- The plus prescription at $p^+ \to 0$ and $p^+ \to k^+$ gives single logarithms $\log(k^+/\Lambda)$ (Step 6). **But** the coefficient of these logarithms is not integrable at large transverse distance, because the Wilson-line factor does not vanish when the soft gluon is far from the target. The true Λ dependence from that region is a *double* logarithm $\log^2(k^+/\Lambda)$ (Step 7).
- The final decomposition, in which every transverse integral converges and all the Λ dependence is explicit, is eq. (33) with eqs. (21), (30), (34), (35).

The UV coefficient, the large-distance limits, the ϵ expansion and the double-log integral were checked numerically against the original six-line bracket.

1 Starting point and notation

The expression to be treated is

$$\begin{aligned}\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{\mathrm{d}p^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}} \\ \times\frac{1}{p^+(y-x)^2+(k^+-p^+)(y-z)^2}\,\frac{(x'-w')^i(z-x)^m}{(x'-w')^2(z-x)^2}\,B^{im} \\ \times\left[U^{ab'}(x')-U^{ab'}(w')\right]W^{af}(x,y,z)\,\rho^{b'}(x')\rho^f(y),\end{aligned} \tag{1}$$

where B^{im} is the six-line square bracket of the original formula. I use

$$\xi = \frac{p^+}{k^+},\quad a=\frac{\Lambda}{k^+},\quad X=y-x,\quad Z=y-z,\quad r=z-x=X-Z,\quad H^i=\frac{(x'-w')^i}{(x'-w')^2}, \tag{2}$$

$$D_\xi = \frac{1}{\xi X^2+(1-\xi)Z^2},\qquad C=\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+},\qquad \langle\Omega\rangle \equiv \left[U^{ab'}(x')-U^{ab'}(w')\right]\Omega^{af}\,\rho^{b'}(x')\rho^f(y). \tag{3}$$

Since $\frac{1}{k^+}\frac{\mathrm{d}p^+}{2\pi}\frac{1}{p^+X^2+(k^+-p^+)Z^2} = \frac{1}{2\pi k^+}\,\mathrm{d}\xi\,D_\xi$, eq. (1) becomes

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = C\int_a^{1-a}\mathrm{d}\xi\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-i\xi k\cdot r}\,D_\xi\,H^iN_\xi^i\,\langle W(x,y,z)\rangle, \tag{4}$$

with $N_\xi^i \equiv r^mB^{im}/r^2$. Contracting each line of the bracket with r^m/r^2 gives

$$N_\xi = \xi(1-\xi)\,n_1 + (1-\xi)\,n_3 + \xi\,n_5 - n_6 - \frac{\xi}{1-\xi}\,n_2 - \frac{1-\xi}{\xi}\,n_4, \tag{5}$$

$$
\begin{aligned}
n_1^i &= d_\perp \frac{r^i\,(X^2-Z^2)}{2r^4}, & n_2^i &= -\frac{X^i}{r^2}\Big[1+\frac{Z\cdot r}{2Z^2}\Big],\\
n_3^i &= \frac{r^i}{r^2}\Big[-\frac{2\,Z\cdot r}{r^2}-\frac{1}{2}+\frac{X\cdot Z}{2X^2}\Big], & n_4^i &= -\frac{Z^i}{r^2}\Big[1-\frac{X\cdot r}{2X^2}\Big],\\
n_5^i &= \frac{r^i}{r^2}\Big[-\frac{2\,X\cdot r}{r^2}+\frac{1}{2}-\frac{X\cdot Z}{2Z^2}\Big],\\
n_6^i &= -\frac{r^i\,(X+Z)\cdot r}{r^4}-\frac{Z^i\,X\cdot r}{2Z^2r^2}+\frac{X^i\,Z\cdot r}{2X^2r^2}.
\end{aligned}
\tag{6}
$$

(Here $X^2-Z^2=r\cdot(X+Z)$ was used.)

2 Step 1: the Wilson-line factor

In the original bracket the terms $+N_cU^{af}(x)-N_cU^{af}(x)$ cancel, so

$$
W^{af}(x,y,z) = -f^{adc}f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(y). \tag{7}
$$

Three limits are needed.

(a) $z\to x$**.** For adjoint Wilson lines $f^{adc}U^{db}(x)U^{ce}(x)=U^{aa'}(x)f^{a'be}$, and $f^{a'be}f^{fbe}=N_c\delta^{a'f}$. Hence

$$
f^{adc}f^{fbe}U^{db}(x)U^{ce}(x)=N_c\,U^{af}(x) \quad\Longrightarrow\quad W(x,y,z)\xrightarrow{z\to x}N_c\big[U(y)-U(x)\big]. \tag{8}
$$

So the piece $-ffU(x)U(z)+N_cU(x)$ vanishes at $z\to x$ and is UV finite; the UV behaviour is that of $N_c[U(y)-U(x)]$ alone.

(b) $x\to\infty$ **or** $z\to\infty$**.** Outside the target $U\to 1$, and

$$
W(x,y,z)\xrightarrow{|x|\to\infty}W_\infty(y,z),\qquad W(x,y,z)\xrightarrow{|z|\to\infty}W_\infty(y,x),\qquad W^{af}_\infty(y,u)\equiv N_cU^{af}(y)-f^{abc}f^{fbe}U^{ce}(u). \tag{9}
$$

W_∞ does *not* vanish. This is what makes Step 7 necessary.

(c) Symmetry. $f^{adc}f^{fbe}U^{db}(x)U^{ce}(z)$ is symmetric under $x\leftrightarrow z$, so $W(x,y,z)=W(z,y,x)$.

3 Step 2: locating the UV divergence

- $x\to y$ and $z\to y$: the integrand behaves as $1/|X|$ or $1/|Z|$, integrable in two dimensions.
- $z\to x$ ($r\to 0$): each n_j is $O(1/r^2)$, so $\int \mathrm{d}^2r/r^2$ diverges logarithmically.

In the limit $r\to 0$ at fixed z: $X\to Z$, $D_\xi\to 1/Z^2$, $e^{-i\xi k\cdot r}\to 1$, and from (6)

$$
n_1\to d_\perp\frac{r^i(Z\cdot r)}{r^4},\quad n_2,\,n_4\to-\frac{Z^i}{r^2},\quad n_3,\,n_5,\,n_6\to-2\,\frac{r^i(Z\cdot r)}{r^4}. \tag{10}
$$

Inserting in (5):

$$
N^i_\xi\to\frac{Z^i}{r^2}\Big[\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big]+\frac{r^i(Z\cdot r)}{r^4}\Big[d_\perp\,\xi(1-\xi)\underbrace{-2(1-\xi)-2\xi+2}_{=0}\Big]. \tag{11}
$$

The angular average in $d_\perp$ dimensions is $r^ir^j/r^2\to\delta^{ij}/d_\perp$, so the explicit $d_\perp$ cancels *exactly* and there is no leftover $O(\epsilon)\times\frac{1}{\epsilon}$ constant:

$$
\boxed{N^i_\xi\xrightarrow{r\to0}\mathcal{P}(\xi)\,\frac{Z^i}{r^2},\qquad \mathcal{P}(\xi)=\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}+\xi(1-\xi).} \tag{12}
$$

With (8) the divergent part of the integrand of (4) is

$$e^{-ik\cdot(w'-z)}\,\frac{H\cdot Z}{Z^2}\,\frac{\mathcal{P}(\xi)}{(x-z)^2}\,N_c\big\langle U(y)-U(z)\big\rangle. \tag{13}$$

4 Step 3: the subtraction term

Subtract and add a term with the same $x\to z$ behaviour that can be integrated over x exactly:

$$\mathcal{S}_\xi = e^{-ik\cdot(w'-z)}\,\frac{H\cdot Z}{Z^2}\,\mathcal{P}(\xi)\,K(x)\,N_c\big\langle U(y)-U(z)\big\rangle, \qquad K(x)\equiv\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}=\frac{X\cdot Z}{X^2r^2}. \tag{14}$$

$K\to 1/(x-z)^2$ for $x\to z$, $K\sim 1/|X|$ for $x\to y$ and $K\sim 1/|x|^3$ for $|x|\to\infty$, so it introduces no new divergence. Then

$$\begin{aligned}\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} &= \underbrace{C\int_a^{1-a}\mathrm{d}\xi\int\mathcal{S}_\xi}_{d^3N^{\mathrm{UV}}/d^3k}\\ &\quad+\underbrace{C\int_a^{1-a}\mathrm{d}\xi\int\Big[e^{-ik\cdot(w'-z)}e^{-i\xi k\cdot r}\,D_\xi\,H\cdot N_\xi\big\langle W\big\rangle-\mathcal{S}_\xi\Big]}_{\text{UV finite, evaluated at } d_\perp=2}.\end{aligned} \tag{15}$$

5 Step 4: dimensional regularisation, $\overline{\mathrm{MS}}$, and the UV logarithms

Transverse integral. Using $X\cdot Z=\frac12[X^2+Z^2-r^2]$,

$$K=\frac12\Big[\frac{1}{r^2}-\frac{1}{X^2}+\frac{Z^2}{X^2r^2}\Big]. \tag{16}$$

The first two terms are scaleless and vanish in dimensional regularisation. For the third, with $d=d_\perp=2-\epsilon$,

$$\int\mathrm{d}^dx\,\frac{1}{X^2\,r^2}=\pi^{d/2}\,\frac{\Gamma\big(\frac{d}{2}-1\big)^2\,\Gamma\big(2-\frac{d}{2}\big)}{\Gamma(d-2)}\,(Z^2)^{\frac{d}{2}-2}. \tag{17}$$

Expanding, $\Gamma(-\frac{\epsilon}{2})^2\Gamma(1+\frac{\epsilon}{2})/\Gamma(-\epsilon)=-\frac{4}{\epsilon}+2\gamma_E+O(\epsilon)$, so

$$\mu^{-\epsilon}\int\mathrm{d}^{2-\epsilon}x\;K(x)=\pi\Big[-\frac{2}{\epsilon}+\gamma_E+\ln\big(\pi Z^2\mu^2\big)\Big]=\pi\Big[-\frac{2}{\epsilon}+2\gamma_E+\ln\frac{(y-z)^2\,\bar\mu^2}{4}\Big], \tag{18}$$

with $\bar\mu^2=4\pi e^{-\gamma_E}\mu^2$. The integral converges for $\epsilon<0$ (i.e. $d_\perp>2$). With the convention $d_\perp=2-\epsilon$ the pole is therefore $-2/\epsilon$; the form $+2/\epsilon$ corresponds to $d_\perp=2+\epsilon$.

Longitudinal integral. $\mathcal{S}_\xi$ depends on ξ only through $\mathcal{P}$:

$$\int_a^{1-a}\mathrm{d}\xi\,\mathcal{P}(\xi)=2\ln\frac{1-a}{a}-\frac{11}{6}+4a-a^2+\frac{2a^3}{3}=2\ln\frac{k^+-\Lambda}{\Lambda}-\frac{11}{6}+O(\Lambda/k^+). \tag{19}$$

UV term. Combining,

$$\begin{aligned}\frac{d^3N^{\mathrm{UV}}}{d^3k} &= \frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\Big[2\ln\frac{k^+-\Lambda}{\Lambda}-\frac{11}{6}\Big]\int_{y,z,x',w'}e^{-ik\cdot(w'-z)}\Big[-\frac{2}{\epsilon}+2\gamma_E+\ln\frac{(y-z)^2\bar\mu^2}{4}\Big]\\ &\quad\times\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\,\big[U^{ab'}(x')-U^{ab'}(w')\big]\big[U^{af}(y)-U^{af}(z)\big]\rho^{b'}(x')\rho^f(y).\end{aligned} \tag{20}$$

In $\overline{\mathrm{MS}}$ the pole is removed:

$$\frac{d^3N^{\rm UV}}{d^3k}\bigg|_{\overline{\rm MS}} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\Big[2\ln\frac{k^+}{\Lambda}-\frac{11}{6}\Big]\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\,\ln\frac{(y-z)^2\bar\mu^2}{c_0^2} \times\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\big[U^{ab'}(x')-U^{ab'}(w')\big]\big[U^{af}(y)-U^{af}(z)\big]\rho^{b'}(x')\rho^f(y), \tag{21}$$

with $c_0 = 2e^{-\gamma_E}$. This is the "UV log" $\ln[(y-z)^2\bar\mu^2]$, multiplied by a rapidity logarithm and the constant $-11/6$.

6 Step 5: the UV-subtracted remainder

The second term of (15) is

$$\frac{d^3N^{\rm rem}}{d^3k} = C\int_a^{1-a}\mathrm{d}\xi\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\Big[e^{-i\xi k\cdot r}D_\xi\,H\cdot N_\xi\langle W\rangle - \mathcal{P}(\xi)\,\frac{H\cdot Z}{Z^2}\,K\,N_c\langle U(y)-U(z)\rangle\Big], \tag{22}$$

with $d_\perp = 2$ in n_1. Two remarks:

- The $1/r^2$ singularities cancel pointwise except for the n_1 term, whose divergence $2\xi(1-\xi)\,r^i(Z\cdot r)/r^4$ cancels against $\xi(1-\xi)Z^i/r^2$ only after the angular integration over the direction of r.
- The remainder still has the endpoint singularities $1/\xi$ and $1/(1-\xi)$, from $-\frac{1-\xi}{\xi}n_4$, $-\frac{\xi}{1-\xi}n_2$ and from $\mathcal{P}(\xi)$ in the subtraction.

7 Step 6: plus prescription (formal)

Write $\frac{1-\xi}{\xi} = \frac{1}{\xi}-1$, $\frac{\xi}{1-\xi} = \frac{1}{\bar\xi}-1$ with $\bar\xi = 1-\xi$, and $\mathcal{P} = \frac{1}{\xi}+\frac{1}{\bar\xi}+\xi(1-\xi)-2$. For any function f,

$$\int_a^{1-a}\frac{\mathrm{d}\xi}{\xi}\,f(\xi) = \int_a^{1-a}\frac{\mathrm{d}\xi}{\xi}\big[f(\xi)-f(0)\big]+f(0)\,\ln\frac{1-a}{a} \;\equiv\; \int\mathrm{d}\xi\Big[\frac{1}{\xi}\Big]_+ f(\xi)+f(0)\ln\frac{1-a}{a}. \tag{23}$$

Endpoint $\xi\to 0$. The singular term is $+\frac{1}{\xi}\,e^{-i\xi k\cdot r}D_\xi\,(H\cdot Z)\frac{1}{r^2}\big[1-\frac{X\cdot r}{2X^2}\big]\langle W\rangle$, and at $\xi = 0$: $e^{-i\xi k\cdot r}D_\xi \to 1/Z^2$.

Endpoint $\xi\to 1$. The singular term is $+\frac{1}{\bar\xi}\,e^{-ik\cdot(w'-x)}e^{i\bar\xi k\cdot r}D_\xi\,(H\cdot X)\frac{1}{r^2}\big[1+\frac{Z\cdot r}{2Z^2}\big]\langle W\rangle$, where $e^{-ik\cdot(w'-z)}e^{-i\xi k\cdot r} = e^{-ik\cdot(w'-x)}e^{i\bar\xi k\cdot r}$ was used; at $\bar\xi = 0$: $D_\xi \to 1/X^2$.

Applying (23) to (22):

$$\begin{aligned}\frac{d^3N^{\rm rem}}{d^3k} \overset{\rm formal}{=}\; & C\int_a^{1-a}\mathrm{d}\xi\int e^{-ik\cdot(w'-z)}\Big[e^{-i\xi k\cdot r}D_\xi\,H\cdot N_\xi^{\rm reg}\langle W\rangle \\ & \qquad -\big(\xi(1-\xi)-2\big)\frac{H\cdot Z}{Z^2}K\,N_c\langle U(y)-U(z)\rangle\Big] \\ & + C\int\mathrm{d}\xi\Big[\frac{1}{\xi}\Big]_+\int e^{-ik\cdot(w'-z)}\,e^{-i\xi k\cdot r}D_\xi\,\frac{H\cdot Z}{r^2}\Big[1-\frac{X\cdot r}{2X^2}\Big]\langle W\rangle \\ & + C\int\mathrm{d}\bar\xi\Big[\frac{1}{\bar\xi}\Big]_+\int e^{-ik\cdot(w'-x)}\,e^{i\bar\xi k\cdot r}D_{1-\bar\xi}\,\frac{H\cdot X}{r^2}\Big[1+\frac{Z\cdot r}{2Z^2}\Big]\langle W\rangle \\ & + C\,\ln\frac{k^+-\Lambda}{\Lambda}\int e^{-ik\cdot(w'-z)}\frac{H\cdot Z}{Z^2}\Big\{\frac{1}{r^2}\Big[1-\frac{X\cdot r}{2X^2}\Big]\langle W\rangle - K\,N_c\langle U(y)-U(z)\rangle\Big\}\end{aligned}$$

$$+\,C\,\ln\frac{k^+-\Lambda}{\Lambda}\int e^{-ik\cdot(w'-x)}\frac{H\cdot X}{X^2}\Big\{\frac{1}{r^2}\Big[1+\frac{Z\cdot r}{2Z^2}\Big]\big\langle W\big\rangle-\frac{X\cdot Z}{Z^2r^2}\,N_c\big\langle U(y)-U(x)\big\rangle\Big\},\quad(24)$$

with $N_\xi^{\rm reg}=\xi(1-\xi)n_1+(1-\xi)n_3+\xi n_5-n_6+n_4+n_2$. In the last line the subtraction term was relabelled $x\leftrightarrow z$ so that the UV cancellation is local. Both curly brackets are UV finite: at $r\to0$ they behave as $\frac{1}{r^2}N_c[U(y)-U(z)]-\frac{1}{r^2}N_c[U(y)-U(z)]$. The constant $-\frac{11}{6}=\int_0^1\mathrm{d}\xi\,[\xi(1-\xi)-2]$ sits in the first line.

Because $W(x,y,z)=W(z,y,x)$, the last two lines are equal after $x\leftrightarrow z$; the single logarithm is therefore $2\ln(k^+/\Lambda)$ times the first curly bracket.

8 Step 7: why the formal result is not yet the answer

The identity (23) needs $f(0)$ to exist. Here $f(0)$ is a transverse integral that *diverges at large distance*, so the single logs in (24) multiply divergent integrals and $\Lambda\to0$ cannot be taken in the plus terms or in the "regular" line.

Large $|x|$ at fixed y,z. From (6), $n_4\to-Z^i/(2x^2)$ and $n_6\to-Z^i/(2Z^2)$, while $W\to W_\infty(y,z)\neq0$. The two contributions combine exactly, for every ξ:

$$D_\xi\Big[\frac{1-\xi}{\xi}\,\frac{Z^i}{2x^2}+\frac{Z^i}{2Z^2}\Big]=\frac{Z^i}{2\xi\,x^2Z^2}\,D_\xi\big[(1-\xi)Z^2+\xi x^2\big]=\frac{Z^i}{2\,\xi\,x^2\,Z^2}.\quad(25)$$

The energy denominator drops out. So the integrand of (4) has the tail

$$\mathcal{A}_0(\xi)=e^{-ik\cdot(w'-z)}\,e^{-i\xi k\cdot r}\,\frac{H\cdot Z}{Z^2}\,\frac{\theta(r^2-Z^2)}{2\,\xi\,r^2}\,\big\langle W_\infty(y,z)\big\rangle,\quad(26)$$

which is cut off only by the phase, at $|x|\sim1/(\xi|k|)$. The x integral gives $\frac{\pi}{\xi}\ln\frac{1}{\xi}$, and the ξ integral then gives $\ln^2(1/a)$. (The remaining tail is $O(\xi^0)/x^2$ and produces only an integrable $\ln\xi$.)

Large $|z|$ at fixed x,y. In the same way $-\frac{\xi}{1-\xi}n_2-n_6$ gives

$$\mathcal{A}_1(\xi)=e^{-ik\cdot(w'-x)}\,e^{i(1-\xi)k\cdot r}\,\frac{H\cdot X}{X^2}\,\frac{\theta(r^2-X^2)}{2\,(1-\xi)\,r^2}\,\big\langle W_\infty(y,x)\big\rangle,\quad(27)$$

which is $\mathcal{A}_0$ with $x\leftrightarrow z$, $\xi\leftrightarrow1-\xi$.

Integrating the tails. With ρ the far coordinate relative to the fixed one, $\kappa=|k|\,|y-z|$ and $\int_s^\infty\frac{\mathrm{d}t}{t}J_0(t)=-\ln\frac{s}{2}-\gamma_E+\int_0^s\frac{\mathrm{d}t}{t}\,[1-J_0(t)]$,

$$\int_a^1\frac{\mathrm{d}\eta}{\eta}\int_{\rho^2>Z^2}\mathrm{d}^2\rho\,\frac{e^{i\eta k\cdot\rho}}{2\rho^2}=\pi\Big[\tfrac12\ln^2\frac{1}{a}-\ln\frac{1}{a}\,\ln\frac{\kappa}{c_0}+h(\kappa)\Big]+O(a),\quad(28)$$

$$h(\kappa)=\int_0^\kappa\frac{\mathrm{d}t}{t}\,\big[1-J_0(t)\big]\ln\frac{\kappa}{t}.\quad(29)$$

Both endpoints give the same contribution (after $x\leftrightarrow z$), so

$$\begin{aligned}\frac{d^3N^{\rm DL}}{d^3k}&\equiv C\int_a^{1-a}\mathrm{d}\xi\int\big[\mathcal{A}_0+\mathcal{A}_1\big]\\&=\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^4}\frac{1}{k^+}\int_{y,z,x',w'}e^{-ik\cdot(w'-z)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\\&\quad\times\Big[\ln^2\frac{k^+}{\Lambda}-2\ln\frac{k^+}{\Lambda}\,\ln\frac{|k|\,|y-z|}{c_0}+2h\big(|k|\,|y-z|\big)\Big]\end{aligned}$$

$$\times \big[U^{ab'}(x') - U^{ab'}(w')\big] W_\infty^{af}(y,z)\, \rho^{b'}(x')\rho^{f}(y). \tag{30}$$

9 Step 8: final result

Subtract $\mathcal{A}_0 + \mathcal{A}_1$ from the remainder (22) and add back (30). Define the transverse-integrated function

$$R(\xi) = \int_{x,y,z,x',w'} \Big\{ e^{-ik\cdot(w'-z)} \Big[e^{-i\xi k\cdot r} D_\xi\, H\cdot N_\xi \big\langle W \big\rangle - \mathcal{P}(\xi) \frac{H\cdot Z}{Z^2} K\, N_c \big\langle U(y) - U(z) \big\rangle \Big] - \mathcal{A}_0(\xi) - \mathcal{A}_1(\xi) \Big\}. \tag{31}$$

Now the endpoint coefficients exist: $\xi R(\xi) \to \mathcal{C}$ for $\xi \to 0$ and $(1-\xi)R(\xi) \to \mathcal{C}$ for $\xi \to 1$ (equal by the $x \leftrightarrow z$ symmetry), with

$$\mathcal{C} = \int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)} \frac{H\cdot Z}{Z^2} \Big\{ \frac{X\cdot(X+Z)}{2X^2 r^2} \big\langle W(x,y,z) \big\rangle - \frac{\theta(r^2 - Z^2)}{2r^2} \big\langle W_\infty(y,z) \big\rangle - \frac{X\cdot Z}{X^2 r^2}\, N_c \big\langle U(y) - U(z) \big\rangle \Big\}, \tag{32}$$

where $\frac{1}{r^2}\big[1 - \frac{X\cdot r}{2X^2}\big] = \frac{X\cdot(X+Z)}{2X^2 r^2}$ was used. The curly bracket is finite at $r \to 0$ ($\frac{1}{2} + \frac{1}{2} - 1 = 0$ times $\frac{1}{r^2} N_c[U(y) - U(z)]$), integrable at $x \to y$, and at large $|x|$ behaves as $\frac{1}{2x^2}\big\langle W - W_\infty \big\rangle + O(1/|x|^3)$, which converges. The plus prescription (23) can now be applied to $R(\xi)$ legitimately.

$$\frac{d^3 N_{2\,\mathrm{NLO}}}{d^3k}\bigg|_{\overline{\mathrm{MS}}} = \frac{d^3 N^{\mathrm{UV}}}{d^3k} + \frac{d^3 N^{\mathrm{DL}}}{d^3k} + \frac{d^3 N^{\mathrm{SL}}}{d^3k} + \frac{d^3 N^{\mathrm{fin}}}{d^3k} \; + \; O(\Lambda/k^+), \tag{33}$$

with d^3N^{UV} in (21), d^3N^{DL} in (30), and

$$\begin{aligned} \frac{d^3 N^{\mathrm{SL}}}{d^3k} &= \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^5} \frac{1}{k^+} \ln\frac{k^+}{\Lambda} \int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)} \frac{(x'-w')\cdot(y-z)}{(x'-w')^2 (y-z)^2} \\ &\quad \times \big[U^{ab'}(x') - U^{ab'}(w')\big] \rho^{b'}(x') \rho^{f}(y) \\ &\quad \times \Big\{ \frac{(y-x)\cdot(2y-x-z)}{2(y-x)^2(z-x)^2}\, W^{af}(x,y,z) - \frac{\theta\big((z-x)^2 - (y-z)^2\big)}{2(z-x)^2}\, W_\infty^{af}(y,z) \\ &\qquad - \frac{(y-x)\cdot(y-z)}{(y-x)^2(z-x)^2}\, N_c\big[U^{af}(y) - U^{af}(z)\big] \Big\}, \end{aligned} \tag{34}$$

$$\frac{d^3 N^{\mathrm{fin}}}{d^3k} = \frac{1}{(2\pi)^3} \frac{g^4}{16\pi^5} \frac{1}{k^+} \int_0^1 \mathrm{d}\xi \Big[R(\xi) - \frac{\mathcal{C}}{\xi} - \frac{\mathcal{C}}{1-\xi} \Big], \tag{35}$$

with $R(\xi)$ in (31) ($d_\perp = 2$) and $\mathcal{C}$ in (32). The subtraction of $\mathcal{C}/\xi$ and $\mathcal{C}/(1-\xi)$ is the plus prescription at the two endpoints. d^3N^{fin} is independent of Λ and of $\bar{\mu}$.

All logarithms collected. With $L \equiv \ln(k^+/\Lambda)$ and $\mathcal{H} \equiv \frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}$:

- UV log: $N_c\big[2L - \frac{11}{6}\big] \ln\frac{(y-z)^2\bar{\mu}^2}{c_0^2}\, \mathcal{H} \big\langle U(y) - U(z) \big\rangle$, prefactor $\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^4 k^+}$.
- Double log: $L^2\, \mathcal{H} \big\langle W_\infty(y,z) \big\rangle$, same prefactor.
- Single logs: $-2L \ln\frac{|k||y-z|}{c_0}\, \mathcal{H} \big\langle W_\infty(y,z) \big\rangle$ (same prefactor) and $2L\,\mathcal{C}$ with prefactor C, eq. (34).

10 Remarks and caveats

1. **Scheme.** Only the divergent x integral was continued to $d_\perp = 2 - \epsilon$. The outer factors ($1/8\pi^4$, the factor $(x' - w')^i/(x' - w')^2$, the numerical $\frac{1}{2}$'s in the bracket) were kept at their two-dimensional values. If they carry ϵ dependence in the derivation of (1), their $O(\epsilon)$ parts times $-2/\epsilon$ add a finite constant multiplying the same structure as (21).

2. **Choice of subtractions.** The UV kernel K and the radius $|y - z|$ in the θ function are conventions. Changing them moves finite terms between $d^3N^{\rm UV}$, $d^3N^{\rm DL}$, $d^3N^{\rm SL}$ and $d^3N^{\rm fin}$; the sum, the $\ln \bar{\mu}^2$ coefficient and the L^2 coefficient do not change.

3. **Order of integration.** In (35) the ξ integral is done after the transverse integrals. At fixed transverse coordinates some terms (e.g. n_6) are not integrable at large distance for $\xi = 0$ exactly; they are for any $\xi > 0$ and give at most integrable $\xi^{-1/2}$ or $\ln \xi$ behaviour.

4. **Assumptions.** $U \to 1$ outside the target, so that $W - W_\infty$ has compact support in the far coordinate; and the outer integrals over y, z, x', w' converge as in the original expression (through the phases and the sources ρ).

5. **The double logarithm.** It comes from a soft gluon ($p^+ \to 0$ or $p^+ \to k^+$) at large transverse distance $|x| \lesssim k^+/(\Lambda|k|)$ and is proportional to $W_\infty \neq 0$. If this diagram class is to be combined with others, that is where the L^2 and the associated $L \ln(|k||y - z|)$ terms should cancel; within this expression alone they are present.

6. **Relation to the "2b" piece.** For $W \to N_c[U(y) - U(x)]$ the UV term is identical to (21), since $-ffU(x)U(z) + N_cU(x)$ vanishes at $z \to x$. The large-distance structure is the same with $W_\infty \to N_c[U(y) - 1]$.